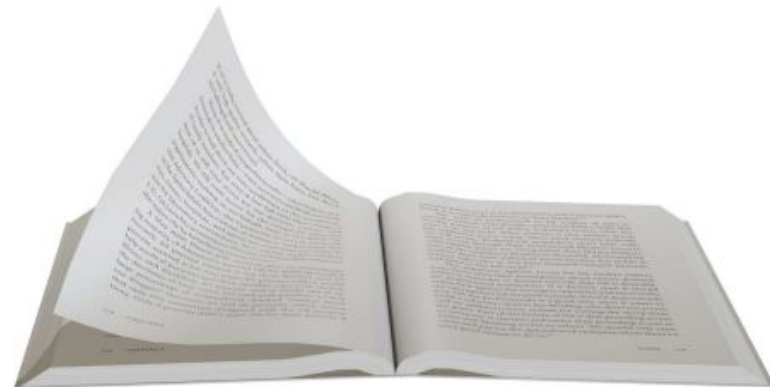
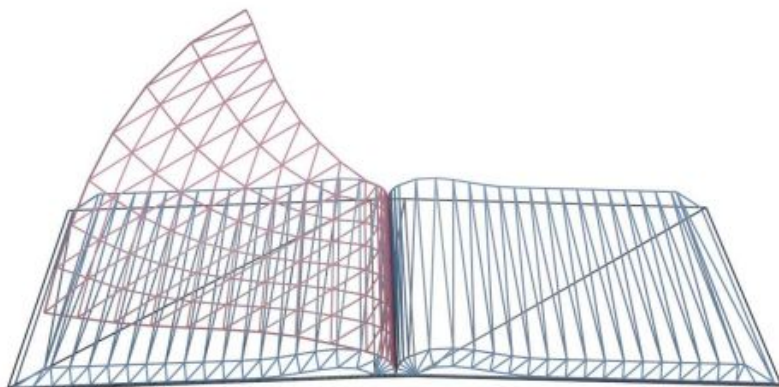


Physically-based Book Simulation with Freeform Developable Surfaces

Thomas Wolf, Victor Cornillère and Olga Sorkine-Hornung
[2021]



Context

Reading from paper compared to screens: A systematic review and meta-analysis [2019]

Reading performance in terms of comprehension and recall appears worse on digital screens compared to physical books.

Wolf et al hypothesis

Slightly better for reading performance

No difference in reading times

Small benefit of paper on metacognition



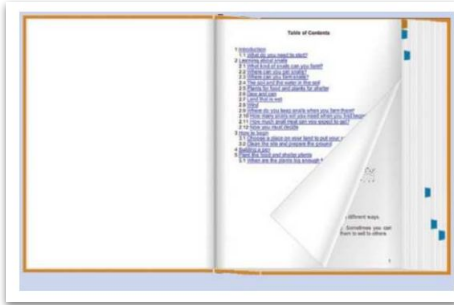
needs of sensorimotor cues ?

missing visual anchors ?

(Just excuses to do something visually nice ?)

Previous Methods

2D animation



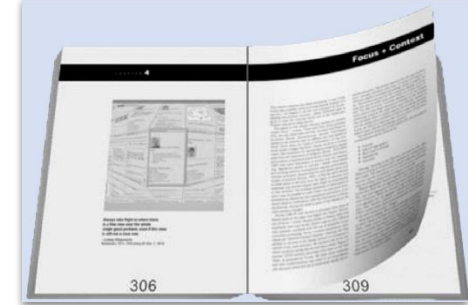
Creating and Reading Realistic Electronic Books
[2009]

Precomputed Simulation (18 x 16 spring grid)



How to turn the page [digital libraries]
[2003]

3D animation



Turning pages of 3D electronic books
[2006]

Really fast

Lack of interactions

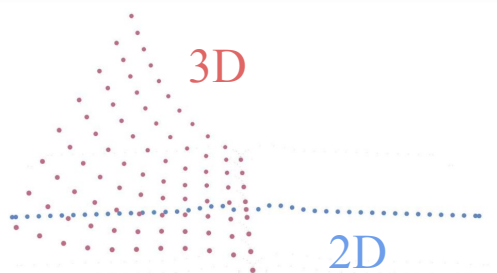
Not visually realistic

No real-time 3D simulation

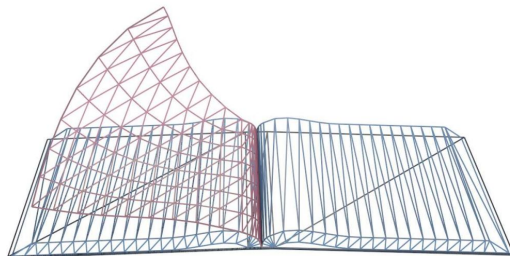
They're not really fair comparisons
(old methods)

Overview

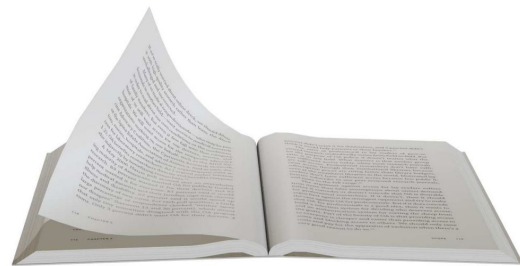
Hybrid 2D/3D simulation



Mesh Generation



Rendering



Constraints

$$C = [C_{\text{DOG}} \quad C_{\text{iso2D}} \quad C_{\text{binding}} \quad C_{\text{fix}}]^T$$

- C_{DOG} : Discrete developability constraints
- C_{iso2D} : Avoid edge length changes for 2D
- C_{binding} : Fix current page to book
- C_{fix} : Fix the book to avoid displacement

Weighted energy potentials

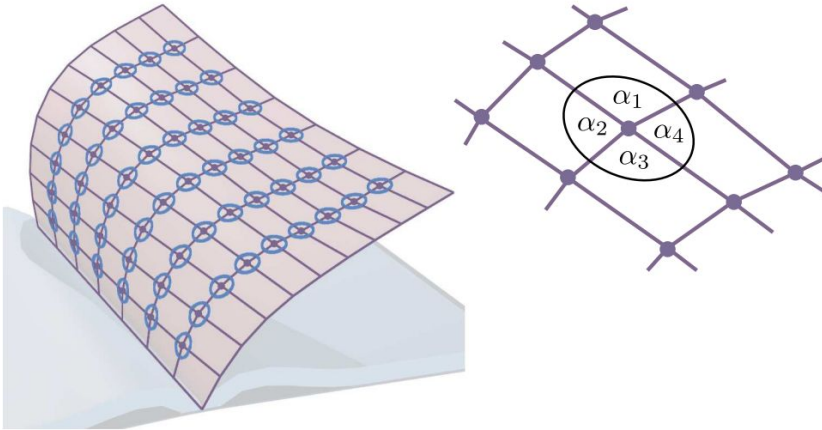
$$V = V_{\text{bending}} + w_{\text{glue}} V_{\text{glue}} + V_{\text{gravity}} + w_{\text{iso3D}} V_{\text{iso3D}} + w_{\text{pos}} V_{\text{pos}}$$

- V_{bending} : 2D and 3D surface smoothing
- V_{glue} : Make the page more or less straight
- V_{gravity} : Well, it's gravity
- V_{iso3D} : Avoid edge length changes for 3D
- V_{pos} : user interactions

DOG Constraint

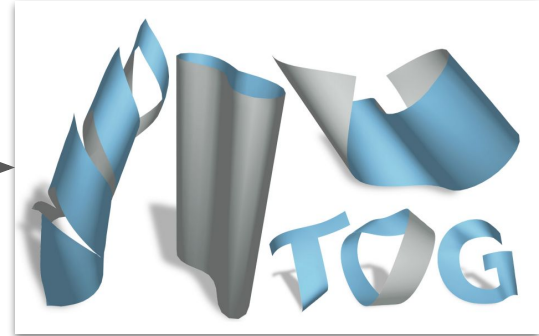
DOG : model the developability of the active page

$$\alpha_1 = \alpha_2 = \alpha_3 = \alpha_4$$



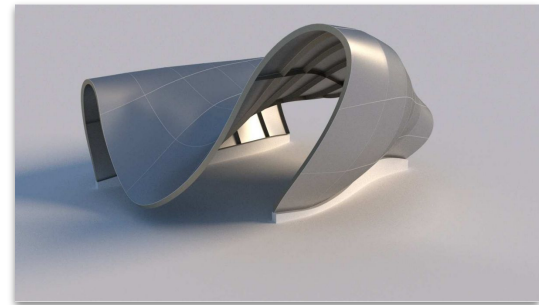
Surface may change its size with deformation !

Add other constraint/energies to avoid unwanted deformation



Discrete Geodesic Nets for Modeling Developable Surfaces [2017]

Possible improvement

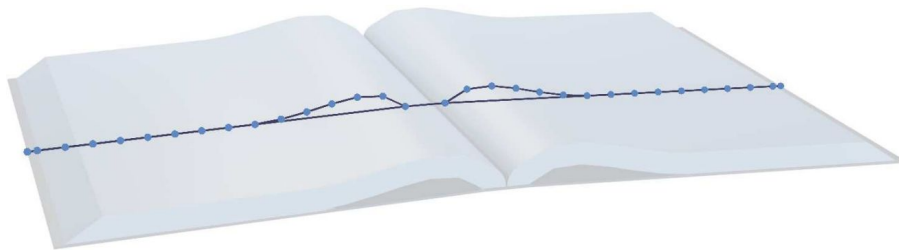


Quad-mesh based isometric mappings and developable surfaces [2020]

Other constraints

Iso2D : Preserve edge length

$$C_{\text{iso2D}(i,j)} = \underbrace{\|x_i - x_j\|^2}_{\text{current}} - \underbrace{\|x_i^0 - x_j^0\|^2}_{\text{init}} = 0$$

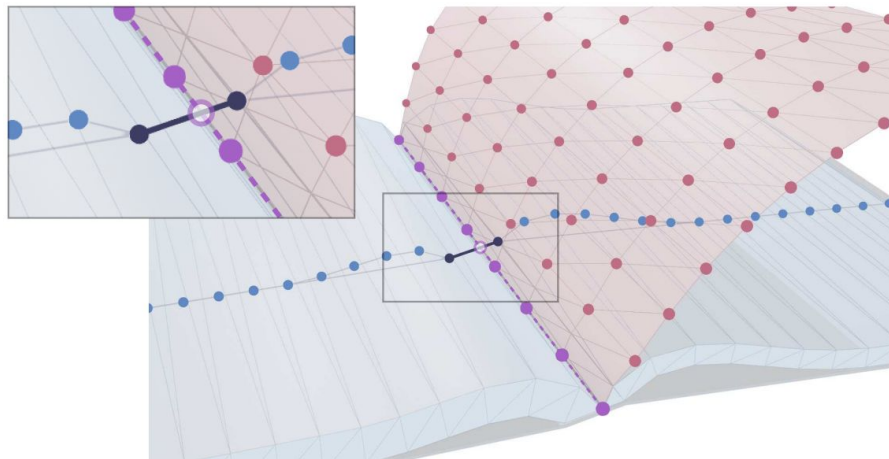


Binding : Fix current page to book

$$C_{\text{binding}(i),X} = x_i^X - \left((1-q)x_{b_1}^X + qx_{b_2}^X \right) = 0$$

$$C_{\text{binding}(i),Y} = x_i^Y - \left((1-q)x_{b_1}^Y + qx_{b_2}^Y \right) = 0$$

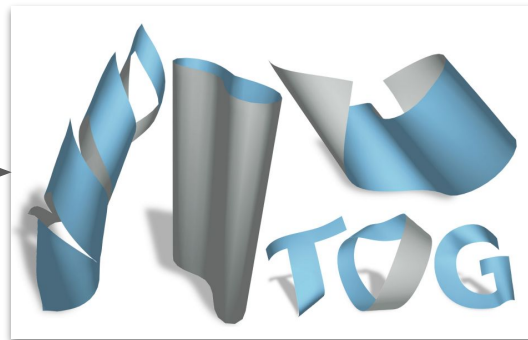
q : location of the page on binding edge



Energy Potentials = “Smooth constraints”

Bending : Smooth shape

$$V_{\text{bending}}(x) = w_{\text{bend2D}} \sum_{i \in \mathcal{S}_{\text{inner-curve}}} \|L(x_i)\|^2 + w_{\text{bend3D}} \sum_{i \in \mathcal{S}_{\text{inner-DOG}}} \|\tilde{L}(x_i)\|^2$$



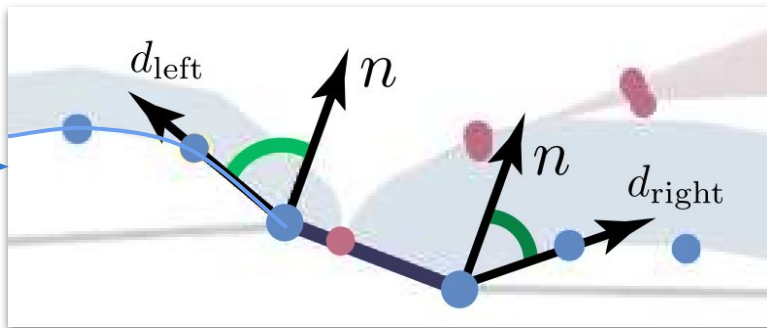
Discrete Geodesic Nets for Modeling Developable Surfaces [2017]

Glue : Hold the pages upright

$$V_{\text{glue}} = (\langle n, d_{\text{left}} \rangle - 1)^2 + (\langle n, d_{\text{right}} \rangle - 1)^2$$

Iso3D : Preserve page edge length

$$V_{\text{iso3D}}(i,j) = \left(\|x_i - x_j\|^2 - \|x_i^0 - x_j^0\|^2 \right)^2$$



Gravity : just gravity

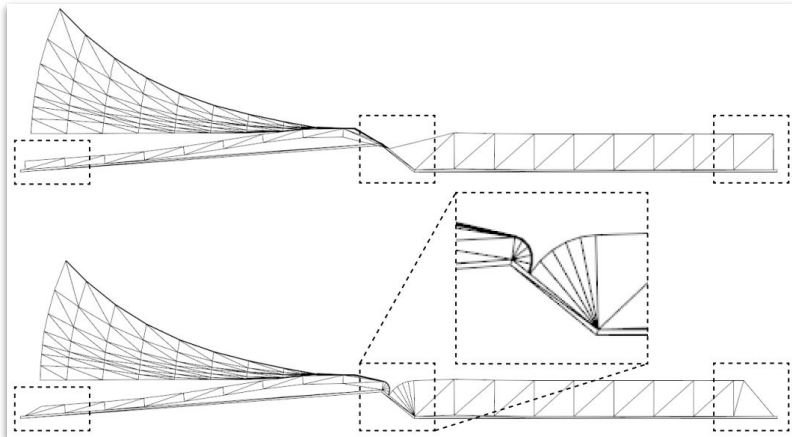
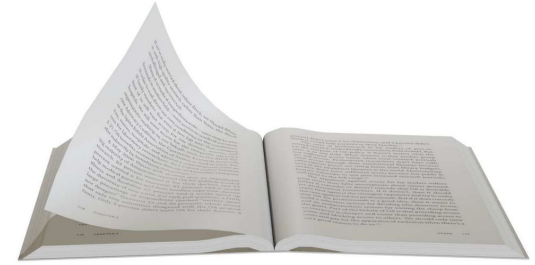
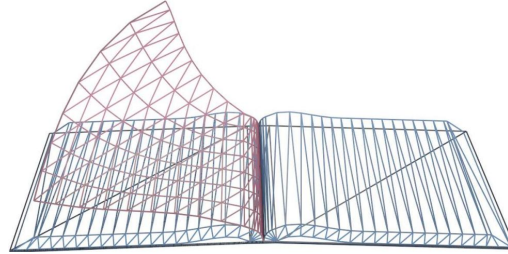
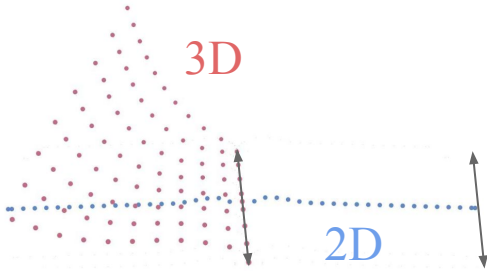
Position : spring like energy

Overview

Hybrid 2D/3D simulation

Mesh Generation

Rendering



Extrusion for the 2D polyline along the binding

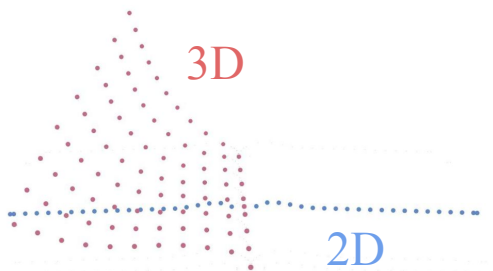
Upsample the 2D polyline close to binding to have a nice curve

Adapt edge length to create a shearing effect

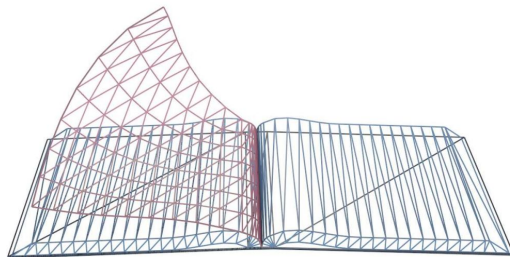
Current page can represent a multiple pages (bigger width)

Overview

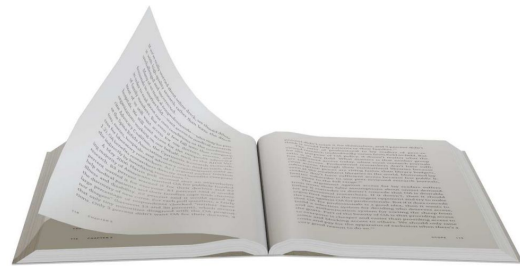
Hybrid 2D/3D simulation



Mesh Generation



Rendering



Classic rendering pipeline, shadow, texture, etc.

Paper is slightly translucent

$$f = k_d f_{\text{Lambert}} + k_s f_{\text{Cook-Torrance}} + k_t f_{\text{Lambert,back}}$$

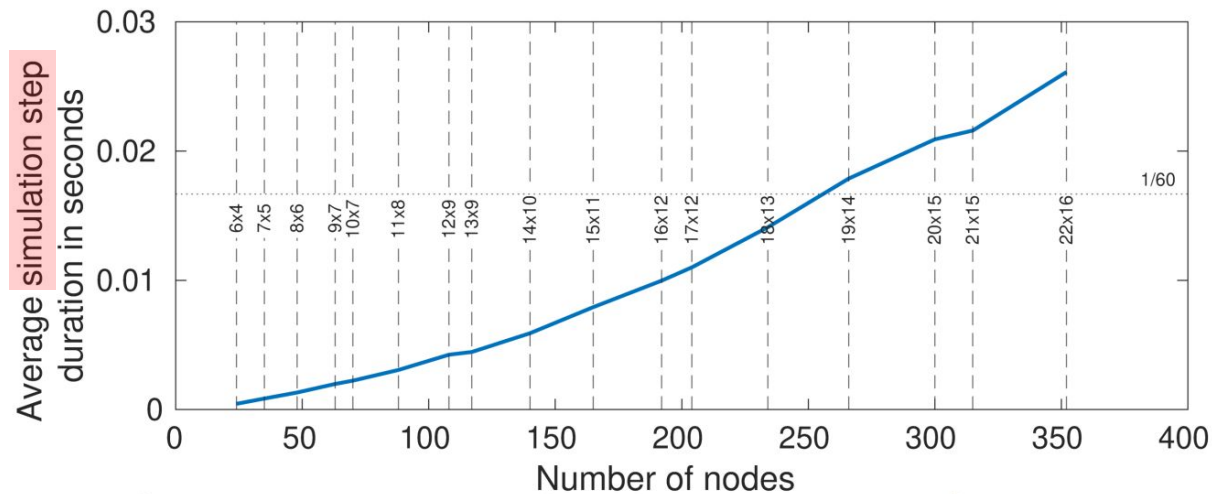
$$k_t = (1 - F_{\text{Schlick,back}}) e^{-\alpha t}$$

$$k_d = (1 - F_{\text{Schlick}}) (1 - e^{-\alpha t})$$

$$k_s = F_{\text{Schlick}} \quad \text{specular reflection}$$

translucency

Performance



6x4

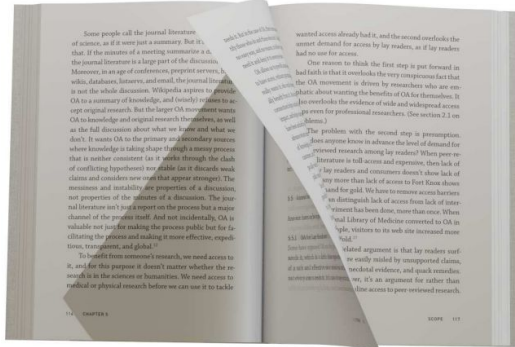


11x8 (ours)

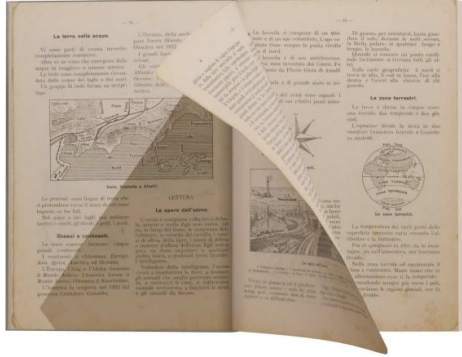


22x16

Final result



new print



vintage book



art catalog



Video